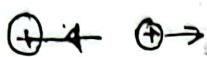
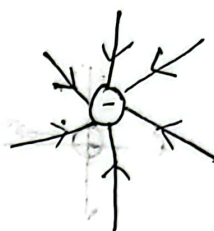
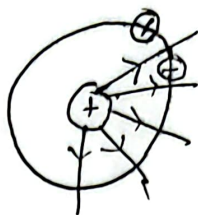


Electric Field: (unit N/C in SI)



$$E = \frac{F}{q_0} = k \frac{q q_0}{q_0 r^2}$$

$$= k \frac{q}{r^2}$$

Electric Field Line

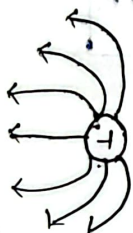
$$q_0 = +1C$$

$$q = ?$$

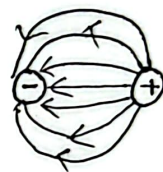
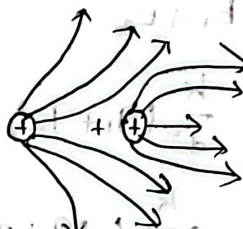
$$r = 2m$$

$$F = 2 \times 10^5 N$$

$$Eq = ?$$



x



show that if is place no place betⁿ two opposite charges where a positive test charge will experience no force

$$q_1 \text{ --- } 5e \text{ --- } q_0 \text{ --- } 2e \text{ --- } q_2$$

$$\text{--- } x=1 \text{ ---}$$

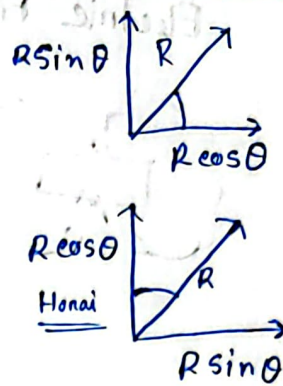
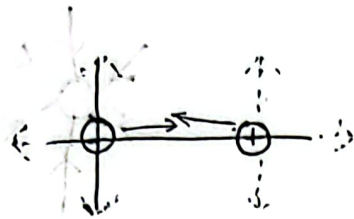
$$r$$

$$\vec{F}_1 + \vec{F}_2 = 0$$

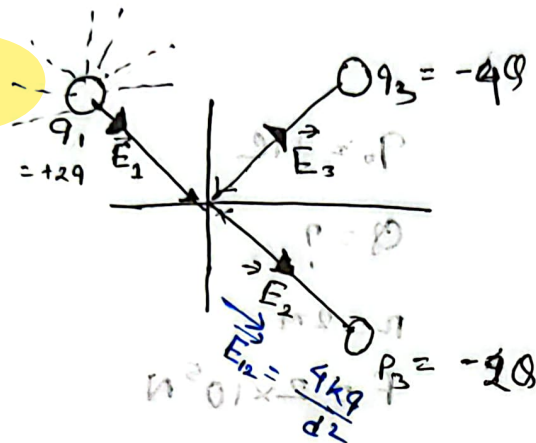
$$\vec{E}_1 + \vec{E}_2 = 0$$

$x = ?$ if imaginary then no force

Electric Field Direction



Question



$$E_1 = \frac{kq_1}{d^2}$$

$$E_2 = \frac{kq_2}{d^2}$$

$$E_3 = \frac{kq_3}{d^2}$$

From the graph

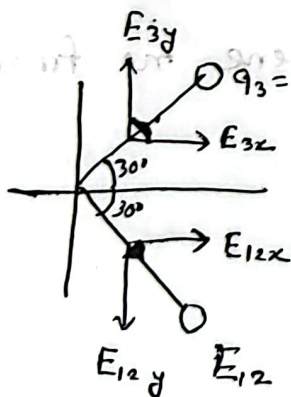
$$\vec{E}_{12} = \vec{E}_1 + \vec{E}_2$$

$$= \frac{k}{d^2} (q_1 + q_2)$$

$$= \frac{k}{d^2} (2q + 2q) \text{ [value calculation: charge (+, -) and distance]}$$

$$\frac{k4q}{d^2}$$

Law of parallelogram



$$E_{3x} = E_3 \cos 30^\circ = \frac{4kq}{d^2} \cos 30^\circ$$

$$E_{3y} = E_3 \sin 30^\circ = \frac{4kq}{d^2} \sin 30^\circ$$

$$E_{12x} = E_{12} \cos 30^\circ = \frac{4kq}{d^2} \cos 30^\circ$$

$$E_{12y} = E_{12} \sin 30^\circ = \frac{4kq}{d^2} \sin 30^\circ$$

Same value
different
direction
'so cancel
out'



$$\therefore E_{x \text{ net}} = E_{3x} + E_{12x}$$

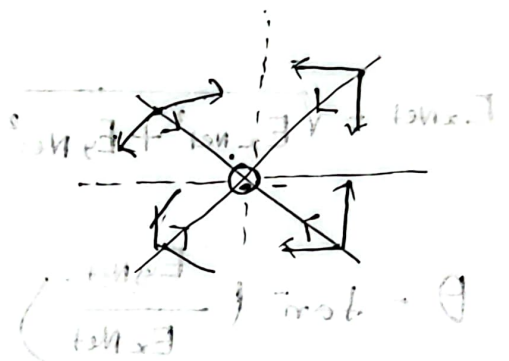
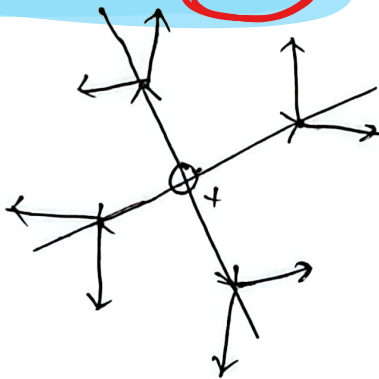
$$= \frac{4kq}{d^2} \cos 30^\circ + \frac{4kq}{d^2} \cos 30^\circ$$

$$= 2 \cdot \left(\frac{4kq}{d^2} \cos 30^\circ \right)$$

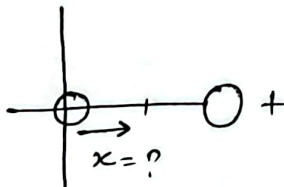


class -15 (Recap)

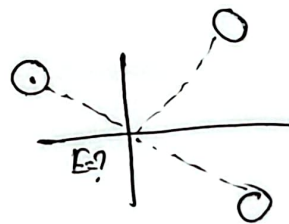
Split



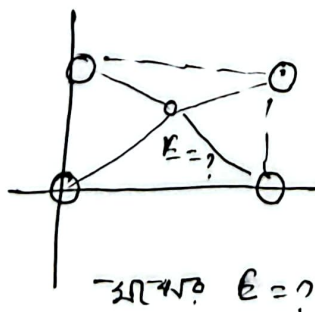
Build picture



what distance if $x=0$?

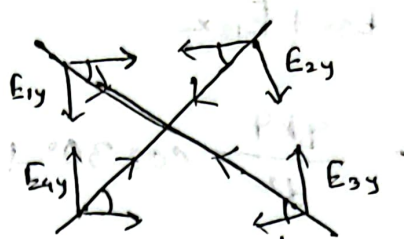
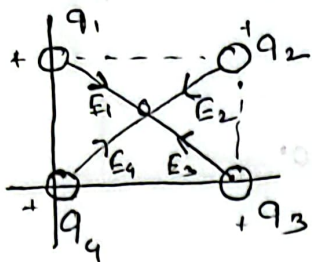


center of $E = ?$



what $E = ?$





$$E_{x\text{net}} = \vec{E}_{1x} + (-\vec{E}_{2x}) + (-\vec{E}_{3x}) + \vec{E}_{4x}$$

$$E_{1x} = E_1 \cos \theta$$

$$E_{1y} = E_1 \sin \theta$$

$$E_{y\text{net}} = -E_{1y} - E_{2y} + E_{3y} + E_{4y}$$

$$E_{\text{net}} = \sqrt{E_{x\text{net}}^2 + E_{y\text{net}}^2}$$

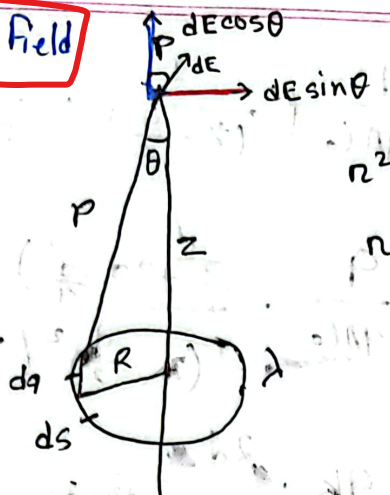
$$\theta = \tan^{-1} \left(\frac{E_{y\text{net}}}{E_{x\text{net}}} \right)$$



C-17

26-11-23

chap-22

Electric Field

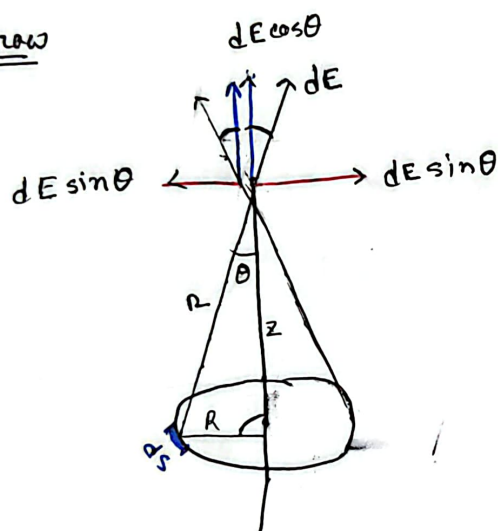
$$r^2 = R^2 + z^2$$

$$r = \sqrt{R^2 + z^2}$$

$$dq = \lambda ds$$

$$dE = \frac{1}{4\pi\epsilon_0} \cdot \frac{dq}{r^2}$$

$$dE = \frac{1}{4\pi\epsilon_0} \cdot \frac{\lambda ds}{(\sqrt{R^2 + z^2})^2}$$

Redraw

$$\cos \theta = \frac{z}{r}$$

$$= \frac{z}{\sqrt{R^2 + z^2}}$$

$$dE = k \frac{ds\lambda}{(\sqrt{R^2 + z^2})^2}$$

$$dE \cos \theta = k \cdot \frac{\lambda ds}{(\sqrt{R^2 + z^2})^2} \cos \theta$$

$$dE \cos \theta = \frac{k ds\lambda}{(R^2 + z^2)} \cdot \frac{z}{\sqrt{R^2 + z^2}}$$

$$a^2 - b^2 = (a+b)(a-b)$$

$$z^2 - \left(\frac{d^2}{4}\right)$$

Electric Dipole



$$E = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{\left(z - \frac{d}{2}\right)^2} - \frac{1}{\left(z + \frac{d}{2}\right)^2} \right)$$

$$= \frac{q}{4\pi\epsilon_0} \cdot \frac{(z^2 + zd + \frac{d^2}{4}) - (z^2 - zd + \frac{d^2}{4})}{\left(z - \frac{d}{2}\right)^2 \left(z + \frac{d}{2}\right)^2}$$

$$= qk \cdot \frac{2dz}{\left(z^2 - \frac{d^2}{4}\right)^2}$$

$$\boxed{z \gg d}$$

$$\boxed{z^2 \gg d^2/4}$$

Problem

$$d = 2H$$

$$E = k \frac{qd}{z^3}$$

$$E = \frac{1}{4\pi\epsilon_0} \frac{q(2H)}{z^3}$$



27-11-23

2000

2000-2001

$\exp(-\beta E) \in [0, 1]$ = probability of state

Problem :

$$H_2O$$

2H number of electron 2

10 ~ 3A-7 ~ 0.8

$$\therefore H_2O \sim H_{n-9} \sim 0.02610$$
$$P = qd \quad \text{Jin 336P}$$

$q = \text{single charge of dipole}$

d = distance between them / dipole.

$$d = 0$$
$$P = 6.2 \times 10^{-30}$$
$$q = 10e = 10 \times 1.6 \times 10^{-19}$$

2. hi fi support degree comparison How eligible is each

There is a 6000 year old history



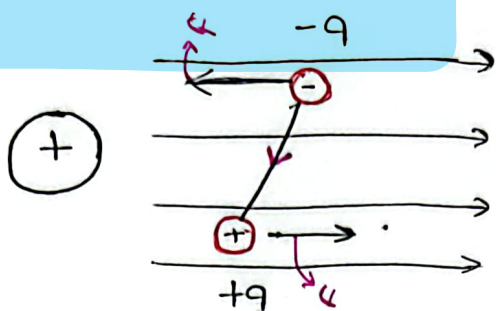
09 2

1000

29. 9

Dipole in Electric field:

→ +



Linear motion force →

angular n → Torque τ

$$\tau = Fd \sin \theta$$

$$\tau = Fd \sin \theta$$

$$F = qE$$

$$= qEd \sin \theta$$

$$p = qd \text{ [dipole moment]}$$

$$= qdE \sin \theta$$

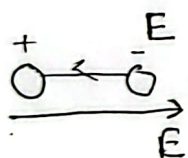
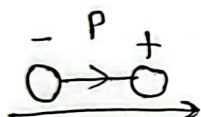
$$\vec{A} \times \vec{B} = AB \sin \theta$$

$$= pE \sin \theta \text{ [} pE \sin \theta \text{ anticlock]} \quad \text{for } p=qd$$

$$\vec{\tau} = \vec{p} \times \vec{E}$$

$$\theta = 0^\circ \text{ or } 180^\circ$$

$$\sin \theta = 0$$



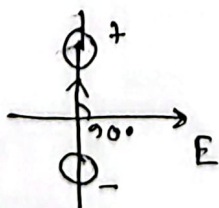
both case τ experience max
at 0° and 180°

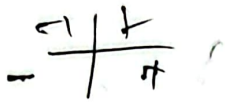
When a dipole will experience max torque if it is placed in a constant electric field?

$$\theta = 90^\circ$$

$$\sin \theta = 1$$

$$\tau = pE$$





Problem

If the molecule is placed in an electric field of 1.5×10^4

what max torque can the field exert on it?

$\Rightarrow$

$$\theta = 90^\circ$$

$$\theta = \sin^{-1}(1) = 90^\circ$$

Work done (τ)

$$dW = F \cdot dr \rightarrow \text{linear}$$

$$dW = \tau d\theta \rightarrow \text{angular}$$

$$\tau = \mu \sin \theta$$

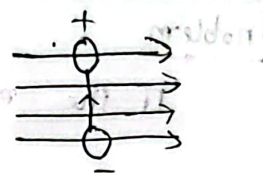
$$\tau = 0.120395 \text{ V}$$

$$\tau = 0.120395 \text{ V}$$

L-20 also

22-5
644P

Electric Field Due to Charge Disk



$$\begin{aligned} U &= -W \\ &= -\int_{90}^{\theta} \tau d\theta \\ &= -\int_{90}^{\theta} -PE \sin \theta \\ &= -PE \cos \theta \Big|_{90}^{\theta} \end{aligned}$$

$$\begin{aligned} U &= -(PE \cos \theta - PE \cos 90^\circ) \quad \cos 90^\circ = 0 \\ &= -PE \cos \theta \end{aligned}$$

$$\vec{A} \cdot \vec{B} = AB \cos \theta$$

$$U = -\vec{P} \cdot \vec{E}$$

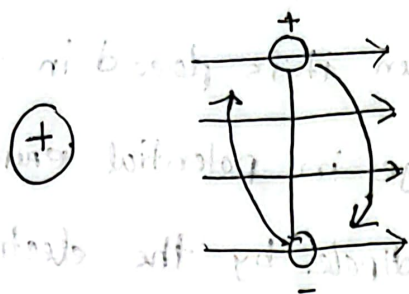
(P) prob 2100

$$\theta = 0^\circ \Rightarrow U = PE \cos 0^\circ = -PE$$

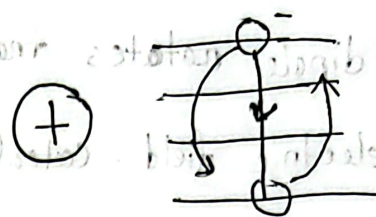
$$U = PE \cos 180^\circ = PE$$

$$U = -PE \cos 90^\circ = 0$$

L-19 → exam 1988

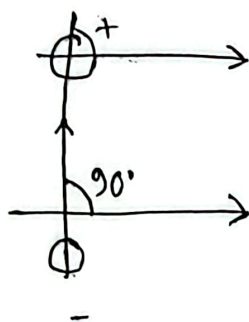


$$\tau = -PE \sin \theta$$



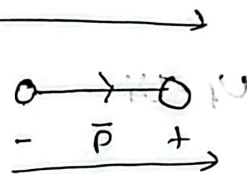
$$\tau = PE \sin \theta$$

Graphically represent max and min torque



$$\theta = 90^\circ$$

$$\theta = 0^\circ$$



$$\theta = 180^\circ$$



$$\tau = 0$$

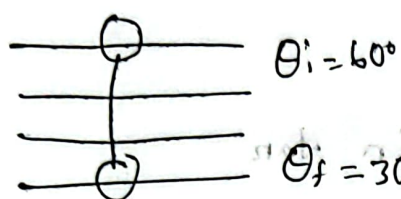
$$\theta = 0^\circ \text{ or } 180^\circ$$

Qn: the electric field is $E=100$ and the dipole moment is $=50$ making an angle $\theta = 60^\circ$. calculate the magnitude of the torque on the dipole

$$\tau = pE \sin \theta \quad [- \text{N}\cdot\text{m}]$$

$$\text{max Torque} = pE \sin 90^\circ = pE$$

a dipole rotates from θ_i to θ_f when it is placed in a electric field. calculate the change in potential energy / calculate the work done on the dipole by the electric field.



$$U_i = -pE \cos \theta_i$$

$$U_f = -pE \cos \theta_f$$

$$U_f - U_i = -pE \cos \theta_f + pE \cos \theta_i$$

$$W = -\Delta U = -(U_f - U_i)$$



work done on the dipole by the electric field

$$W = -\Delta U = -(U_f - U_i)$$

$$= -(-pE \cos \theta_f + pE \cos \theta_i)$$

work done on the dipole by an external agent

$$W_a = -W$$

$$= (U_f - U_i)$$

$$= (-pE \cos \theta_f + pE \cos \theta_i)$$

$$U_f = -50j$$

$$U_i = -75j$$

$$\text{respect to EF} \rightarrow W = -(-50 - (-75)) \\ = -25j$$

$$\text{respect to ex. A} \rightarrow W = (-50 + 75) \\ = 25j$$

External agent

$$\begin{array}{c} \longrightarrow \\ \longrightarrow \end{array} \quad \theta_i = 0^\circ$$

$$\begin{array}{c} \curvearrowright \\ \ominus \longrightarrow \oplus \\ \curvearrowleft \end{array} \quad \theta_f = 180^\circ$$

P

$$W = U_f - U_i$$

$$= -PE \cos 180^\circ - (-PE \cos 0^\circ)$$

$$= +PE + PE$$

$$= 2PE$$